HTML Fundamentals-day1

1.1 Metadata Elements

Metadata Overview

- Stored inside the `<head>` section.
- Not displayed on the webpage.
- Provides information to browsers, search engines, and services.

Meta Tag (<meta>)

Purpose

- Defines metadata about an HTML document.
- Self-closing (void) element.

Character Encoding

```
<meta charset="UTF-8">
```

Why Important

- Supports characters from all languages.
- Prevents display issues with symbols such as:
 - ?
 - □
 - Ø

Keywords

```
<meta name="keywords" content="HTML, CSS, web development, tutorial">
```

Purpose

- Lists keywords related to the page.
- Multiple keywords separated by commas.
- Mostly ignored by modern search engines.

Description

```
<meta name="description" content="Learn HTML and CSS Fundamentals for the ITI certification exam.">
```

Purpose

- Shown in search engine results.
- Improves click-through rates.
- Should be unique for every page.
- Recommended length: under 160 characters.

Viewport

```
<meta name="viewport" content="width=device-width, initial-scale=1.0">
```

Purpose

- Makes websites responsive.

Components

- width=device-width
 - Matches page width to device width.
- initial-scale=1.0
 - Sets initial zoom level.

Without Viewport

- Mobile browsers display pages like desktop websites.

Link Tag (<link>)

Purpose

- Connects to HTML documents with external resources.
- Self-closing element.

Common Uses

External CSS

```
<link rel="stylesheet" href="styles.css">
```

Favicon

```
<link rel="icon" href="favicon.ico" type="image/x-icon">
```

Google Fonts

```
<link rel="preconnect" href="https://fonts.googleapis.com">
```

Important Attributes

rel

- Defines relationship.
- Examples:
 - stylesheet
 - icon
 - preconnect

href

- Location of resource.

type

- MIME type of resource.

Style Tag (<style>)

Purpose

- Internal CSS.

Example

```
<style>
  body {
    font-family: Arial, sans-serif;
    background-color: #f0f0f0;
  }
  h1 {
    color: navy;
  }
</style>
```

Key Points

- Affects only the current page.
- Useful for page-specific styling.
- Higher priority than external CSS when specificity is equal.

Script Tag (<script>)

Purpose

- Adds JavaScript to a webpage.

Internal JavaScript

```
<script>
  alert("Hello, World!");
</script>
```

External JavaScript

```
<script src="app.js"></script>
```

Defer

```
<script src="app.js" defer></script>
```

Behavior

- Downloads in background.
- Runs after HTML parsing.

Async

```
<script src="analytics.js" async></script>
```

Behavior

- Downloads in background.
- Runs immediately when ready.
- Execution order not guaranteed.

Best Practice

- Place scripts before `</body>`.
- Or use defer.

Noscript Tag (<noscript>)

Purpose

- Displays content when JavaScript is disabled.

Example

```
<noscript>
  <p>Please enable JavaScript.</p>
</noscript>
```

Benefits

- Accessibility
- Graceful degradation

Metadata Elements Summary

Meta

- Page information
- Self-closing

Link

- External resources
- Self-closing

Style

- Internal CSS
- Requires closing tag

Script

- JavaScript
- Requires closing tag

Noscript

- Fallback content
- Requires closing tag

1.2 Well-Formed Page Markup

DOCTYPE Declaration

```
<!DOCTYPE html>
```

Purpose

- Declares HTML5.
- Must be first line of document.
- Prevents Quirks Mode.

HTML Element

```
<html lang="en">
</html>
```

Purpose

- Root element.
- Contains all page content.

lang Attribute

- Specifies page language.
- Helps accessibility and SEO.

Head Element

```
<head>
</head>
```

Contains

- title
- meta
- link
- style
- script

Not Visible to Users

Body Element

```
<body>
</body>
```

Contains

- Headings
- Paragraphs
- Images
- Forms
- Visible webpage content

Rule

- Only one body element per page.

Complete HTML Structure

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <meta name="description" content="A sample HTML page.">
    <title>Page Title</title>
    <link rel="stylesheet" href="styles.css">
  </head>
  <body>
    <h1>Hello World</h1>
    <p>This is my first webpage.</p>
    <script src="app.js" defer></script>
  </body>
</html>
```

Proper Syntax Rules

Rule 1: Correct Nesting

Wrong

```
<b><i>Text</b></i>
```

Correct

```
<b><i>Text</i></b>
```

Rule 2: Quote Attributes

Wrong

```
<img src=photo.jpg>
```

Correct

```
<img src="photo.jpg" alt="A photo">
```

Rule 3: Use Lowercase Tags

Preferred

```
<p>Paragraph</p>
```

Rule 4: Close Tags Properly

Void Elements (No Closing Tag)

- br
- hr
- img
- input
- meta
- link

Regular Elements (Need Closing Tag)

- p
- div
- span
- H

HTML Entities

Why Use Entities?

- Display reserved characters safely.

Ampersand

```
&amp;
```

Output: &

Less Than

```
&lt;
```

Output: <

Greater Than

```
&gt;
```

Output: >

Double Quote

```
&quot;
```

Output: "

Apostrophe

```
&apos;
```

Output: '

Non-Breaking Space

```
 
```

Copyright

```
&copy;
```

Output: ©

Registered Trademark

```
&reg;
```

Output: ®

Trademark

```
&trade;
```

Output: ™

Euro

```
&euro;
```

Output: €

Indian Rupee

```
&#8377;
```

Output: ₹

Right Arrow

```
&rarr;
```

Output: →

Key Takeaways

HTML Document Structure

DOCTYPE | HTML | HEAD | BODY

Essential Metadata

- charset
- viewport
- title
- description

Good HTML Practices

- Proper nesting
- Quote attributes
- Use lowercase tags
- Close regular elements

Remember

- Head = Metadata
- Body = Visible Content
- Entities = Special Characters